EXPERIMENT NO. 06

AIM: Create an app to fetch and display data from a REST API.

THEORY:

- **REST API**
  - Web service that provides data over HTTP
  - Uses methods like GET to retrieve data
- **JSON Data Format**
  - Lightweight format for data exchange
  - Parsed using `JSONObject` and `JSONArray`
- **Networking in Android**
  - Data fetched using URL connection
  - Must run on background thread to avoid UI blocking
- **RecyclerView**
  - Efficiently displays large datasets
  - Reuses views for better performance
- **Adapter Pattern**
  - Binds data to RecyclerView items
  - Converts raw data into UI components
- **User Experience Enhancements**
  - ProgressBar shows loading state
  - SwipeRefreshLayout enables pull-to-refresh

In AndroidManifest

```
<uses-permission android:name="android.permission.INTERNET"/>
```

MainActivity.kt

```
package com.example.myapplication

import android.os.Bundle
import android.view.View
import android.widget.Button
import android.widget.ProgressBar
import
androidx.appcompat.app.AppCompatActi
vity
import
androidx.recyclerview.widget.LinearL
ayoutManager
import
androidx.recyclerview.widget.Recycle
rView
import
androidx.swiperefreshlayout.widget.S
wipeRefreshLayout
import org.json.JSONArray
import java.net.URL

class MainActivity :
AppCompatActivity() {
   private lateinit var
recyclerView: RecyclerView
   private lateinit var progressBar:
ProgressBar
   private lateinit var adapter:
MyAdapter
   private lateinit var
swipeRefresh: SwipeRefreshLayout

   private val dataList =
mutableListOf<Todo>()

   override fun
onCreate(savedInstanceState:
Bundle?) {

super.onCreate(savedInstanceState)

setContentView(R.layout.activity_mai
n)
```

```
        val btnFetch =
findViewById<Button>(R.id.btnFetch)
        recyclerView =
findViewById(R.id.recyclerView)
        progressBar =
findViewById(R.id.progressBar)
        swipeRefresh =
findViewById(R.id.swipeRefresh)

        recyclerView.layoutManager =
LinearLayoutManager(this)
        adapter = MyAdapter(dataList)
        recyclerView.adapter =
adapter

        btnFetch.setOnClickListener {
            fetchData()
        }


swipeRefresh.setOnRefreshListener {
            fetchData()
            swipeRefresh.isRefreshing
= false
        }
    }

    private fun fetchData() {
        progressBar.visibility =
View.VISIBLE
        dataList.clear()

        Thread {
            try {
                val url =
"https://jsonplaceholder.typicode.co
m/todos"
                val response =
URL(url).readText()

                val jsonArray =
JSONArray(response)

                for (i in 0 until 10)
{
                    val obj =
jsonArray.getJSONObject(i)
                    val title =
obj.getString("title")
                    val completed =
obj.getBoolean("completed")


dataList.add(Todo(title, completed))
                }

                runOnUiThread {

progressBar.visibility = View.GONE

adapter.notifyDataSetChanged()
                }

            } catch (e: Exception) {
                runOnUiThread {

progressBar.visibility = View.GONE
                }
            }
        }.start()
    }
}
```

MyAdapter.kt

```
package com.example.myapplication

import android.view.LayoutInflater
import android.view.ViewGroup
import android.widget.CheckBox
import android.widget.TextView
import android.widget.Toast
import
androidx.recyclerview.widget.Recycle
rView

class MyAdapter(private val
dataList: List<Todo>) :

RecyclerView.Adapter<MyAdapter.ViewH
older>() {

    class ViewHolder(parent:
ViewGroup) :
RecyclerView.ViewHolder(

LayoutInflater.from(parent.context)
```

```
.inflate(R.layout.item_layout,
parent, false)
   ) {
       val tvItem: TextView =
itemView.findViewById(R.id.tvItem)
       val checkBox: CheckBox =
itemView.findViewById(R.id.checkBox)
   }

   override fun
onCreateViewHolder(parent:
ViewGroup, viewType: Int):
ViewHolder {
       return ViewHolder(parent)
   }

   override fun
onBindViewHolder(holder: ViewHolder,
position: Int) {
       val item = dataList[position]

       holder.tvItem.text =
item.title
       holder.checkBox.isChecked =
item.completed
       // Click event

holder.itemView.setOnClickListener {
           Toast.makeText(

holder.itemView.context,
               "Clicked:
${item.title}",
               Toast.LENGTH_SHORT
           ).show()
       }
       // Change color if completed
       if (item.completed) {
           holder.tvItem.alpha =
0.5f
       } else {
           holder.tvItem.alpha =
1.0f
       }
   }

   override fun getItemCount(): Int
= dataList.size
}
```

Todo.kt

```
package com.example.myapplication

data class Todo(
   val title: String,
   val completed: Boolean
)
```

item_layout.xml

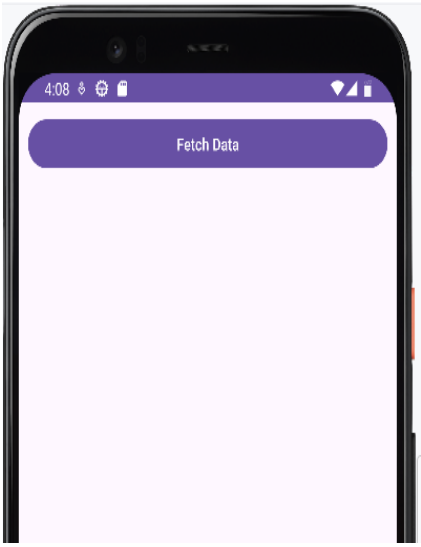
```
<?xml version="1.0"
encoding="utf-8"?>
<androidx.cardview.widget.CardView
xmlns:android="http://schemas.androi
d.com/apk/res/android"

android:layout_width="match_parent"

android:layout_height="wrap_content"
   android:layout_margin="8dp"
   android:elevation="6dp"
   android:padding="8dp"
   app:cardCornerRadius="12dp"

xmlns:app="http://schemas.android.co
m/apk/res-auto">
   <LinearLayout

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:orientation="horizontal"
       android:padding="12dp">

       <CheckBox

android:id="@+id/checkBox"

android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
/>

       <TextView
           android:id="@+id/tvItem"

android:layout_width="0dp"

android:layout_height="wrap_content"
```

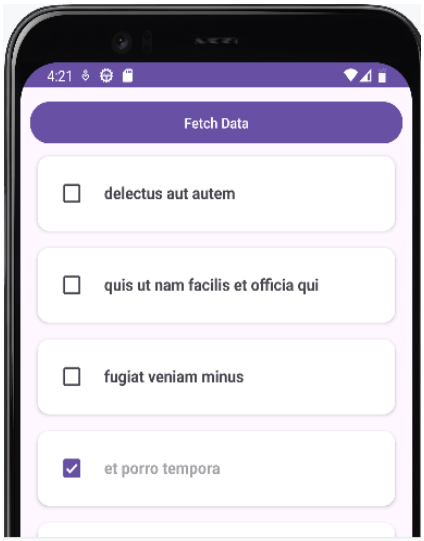
Activity_main.xml

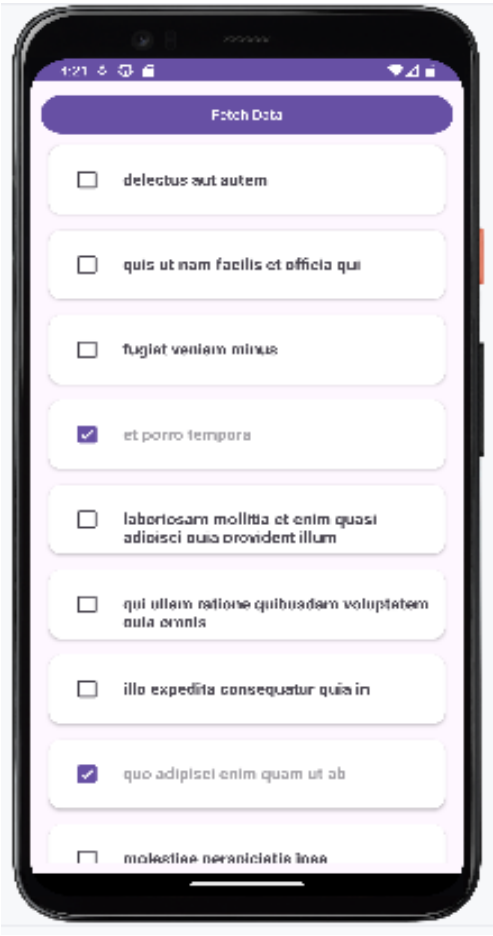
```
<?xml version="1.0"
encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.androi
d.com/apk/res/android"

android:layout_width="match_parent"

android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="10dp">

   <Button
       android:id="@+id/btnFetch"

android:layout_width="match_parent"

android:layout_height="wrap_content"
       android:text="Fetch Data"/>

   <ProgressBar
       android:id="@+id/progressBar"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:layout_gravity="center"
       android:visibility="gone"
            android:layout_weight="1"
            android:textSize="16sp"
            android:textStyle="bold"

android:paddingStart="10dp"/>

   </LinearLayout>

</androidx.cardview.widget.CardView>



android:layout_marginTop="10dp"/>


<androidx.swiperefreshlayout.widget.
SwipeRefreshLayout

android:id="@+id/swipeRefresh"

android:layout_width="match_parent"

android:layout_height="match_parent"
>


<androidx.recyclerview.widget.Recycl
erView

android:id="@+id/recyclerView"

android:layout_width="match_parent"

android:layout_height="match_parent"
/>


</androidx.swiperefreshlayout.widget
.SwipeRefreshLayout>

</LinearLayout>
```

Home screen



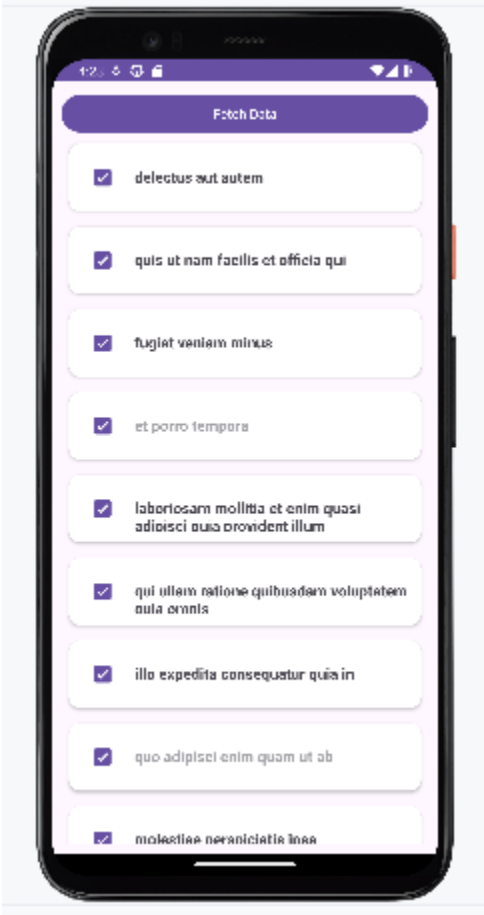
After clicking fetch data



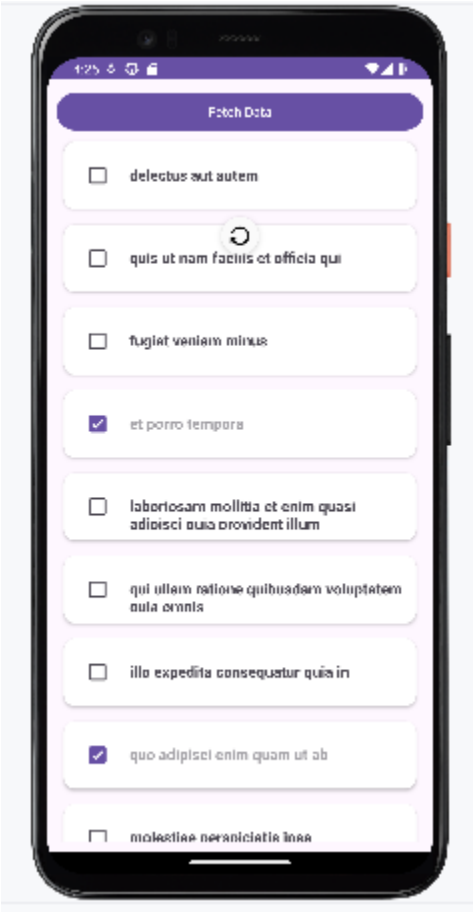


User interaction: After clicking on a list item, toast message comes up

Can check all checkboxes



Swipe down to refresh



CONCLUSION:

This experiment demonstrates how to fetch, parse, and display data from a REST API using RecyclerView, enabling dynamic data handling and improved user experience in Android applications.